Metabolic syndrome derived at age 53 years (1999)

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Metabolic syndrome derived variables at age 53.

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Document details

Categories of variables	Cardiovascular health
Summary of work undertaken	Derivation and labelling of Metabolic syndrome variables: **DIABMS** - Diabetes status at age 53y (recoded for metabolic syndrome) **WCBIN99** - High waist circumference **SHYP992M** - High systolic blood pressure **DHYP992M** - High diastolic blood pressure **HYP992M** - Hypertension **HBABIN99** - High level of glycated haemoglobin (using quartile as cut-off) **HBABIN99M** - High level of glycated haemoglobin (using quartile as cut-off) or medication for diabetes at age 53y **HDLBIN99** - Low level of high density lipoprotein at age 53y **TRIGBIN99** - High level of triglyceride at age 53y **MSSCORE99** - Number of metabolic syndrome symptoms at age 53y **METABOL99** - Metabolic syndrome at age 53y **TRIGBIN99X** - High level of triglyceride at age 53y (quartile as a cut-off) **MSSCORE99X** - Number of metabolic syndrome symptoms at age 53y (TRIGBINX) **METABOL99X** - Metabolic syndrome at age 53y (TRIGBINX)
Names of source variables	sbp992, dbp992, hba99n, chol99n, trig99n, hdl99n, ldl99n, abc, hipc, bmi99u, b02299a, b02499a, b02599a, b026299a, b06199a, b212099a, diabty, abcc, hipcc, whr99, medbp99
Output variables	diabms, wcbin99, shyp992m, dhyp992m, hyp992m, hbabin99, hbabin99m, hdlbin99, trigbin99, msscore99, metabol99, trigbin99x, msscore99x, metabol99x
Papers using these variables	Gaysina, D, Pierce, M, Richards, M, Hotopf, M, Kuh, D, Hardy, R. Association between adolescent emotional problems and metabolic syndrome: the modifying effect of C-reactive protein gene polymorphisms. Brain, Behavior, and Immunity 2011; 25: 750-758.

Overview

Metabolic syndrome derived variables at age 53 (1999).

Metabolic syndrome definition

The metabolic syndrome and its components were defined using cut points recommended by ATPIII8 (2001). We modified the ATPIII8 definition to include glycated haemoglobin (HbA1c) instead of fasting plasma glucose, data for which were unavailable. Participants were classified as having the metabolic syndrome if they met any 3 of the following criteria: 1) Waist circumference > 102 cm for men or > 88 cm for women, 2) Triglyceride level ≥ 1.7 mmol/L (150 mg/dL), Since the triglyceride level was measured using non-fasting blood, we additionally defined another threshold - in the top gender-specific quarter of the distribution (≥ 3.0 mmol/L for men and ≥ 2.2

for women). 3) HDL cholesterol level < 1.036 mmol/L for men or <1.295 mmol/L for women, 4) Blood pressure level: either SBP ≥ 130 or DBP ≥ 85 mm Hg. Participants using antihypertensive medications were considered as meeting high blood pressure. 5) HbA1c level in the top gender-specific quarter of the distribution (> 5.8% among both men and women). Participants using medications for diabetes were considered as meeting HbA1c criteria.

Raw variables and variables derived previously

SBP992 – systolic blood pressure at age 53; **DBP992** – diastolic blood pressure at age 53;

- Missing for all reasons is coded to 9999 for all variables.
- These are second measures of the blood pressure: All are in mmHg.
- These variables can be picked up from: $der/final/cvd/bp

HBA99N – glycated haemoglobin (%) at age 53; **CHOL99N** – total cholesterol (mmol/l) at age 53; **TRIG99N** – triglycerides (mmol/l) at age 53; **HDL99N** – high density lipoprotein cholesterol (mmol/l) at age 53; Note: HDL values of 8 are excluded; **LDL99N** – low density lipoprotein cholesterol (mmol/l) at age 53;

- Missing values are then 77 (no blood taken?), 98 (different missing data), 99 (no interview)
- The location of these variables is $der/final/cvd/bloods

ABC – abdominal (waist) circumference (cm) at age 53; **HIPC** – hips circumference (cm) at age 53, cm

- These original variables were recoded to ABC99C and HIPC99C with missing: 999 (no interview, based on Q99A answer) and 988 (different missing data). These variables are kept together in the data file.
- The original variables can be picked up from $xlb/capi99a/c.sv
- The recoded variables can be picked up from $der/final/waisthip/

Q99A – whether the Survey member was interviewed in 1999: 0=no, 1=yes **BMI99U** – BMI (body mass index) in 1999 in kg/(m2); can be picked up from the derived library $der\final\htwtbmi\d080422.sv **B02299A**, **B02499A**, **B02599A**, **B026299A** – antihypertensive medication: diuretics, beta blockers, RA system drugs+others, calcium channel blockers, correspondingly; **B06199A** – medication for diabetes; **B212099A** – medication for dislipidemia.

These variables can be picked up from $xlb/bnf99a/d020619.sv

DIABTY - Type of diabetes 1=Type 1 - diagnosed younger age (28 or less in this group) and on insulin since diagnosis 2=Type 2 - diagnosed at older age (30 or more in this group) and only ever on diet or OHA, or insulin added more than 2 years after diagnosis 3=GDM- diagnosed in pregnancy and no further record of treatment for diabetes 4=Unknown type 5=Unsure ? type 1 / type 2 – see notes in document: males_and_females_diagnosed_by_1999 6=Unsure ? type 2 / type 1 – see notes in document: males_and_females_diagnosed_by_1999 88 =No diabetes 99=Missing This variable can be picked up from the derived library: $is/diabetes/derived/d090305.sv

MEDBP99 – antihypertensive medication use at age 53 0 = no antihypertensive medication use 1 = antihypertensive medication use 999 = missing

WHR99 – waist-hip ratio at age 53 (can be useful for WHO metabolic syndrome definition) 999 = missing

Derived variables

DIABMS

Purpose : Diabetes status at age 53yrs (recoded for metabolic syndrome) Year : 1999

1. Definition

0 No (including those with gestational diabetes)
1 Yes (except those with gestational diabetes)
999 Missing

2. Specification

COMPUTE DIABMS=999. DO IF (DIABTY = 1) OR (DIABTY = 2) OR (DIABTY = 4) OR (DIABTY = 5) OR (DIABTY = 6). COMPUTE DIABMS=1. ELSE IF (DIABTY = 3) OR (DIABTY = 88). COMPUTE DIABMS=0. ELSE IF (DIABTY = 99). COMPUTE DIABMS=999.

WCBIN99

Purpose : High waist circumference at age 53yrs Year : 1999

1. Definition

0 No
1 Yes 999 Missing

2. Specification

RECODE ABC (ELSE=COPY) INTO ABCC. RECODE ABCC (SYSMIS=988). IF (ABC = .0) ABCC=988. IF (ABC = .9) ABCC=988. IF (ABC = 999.9) ABCC=988. RECODE HIPC (ELSE=COPY) INTO HIPCC. RECODE HIPCC (SYSMIS=988). IF (HIPC = .0) HIPCC=988. IF (HIPC = 999.0) HIPCC=988. IF (HIPC = 999.9) HIPCC=988. DO IF Q99A=0. COMPUTE ABCC=999. COMPUTE HIPCC=999. END IF. COMPUTE ABCC99=ABCC. RECODE ABCC99 (988=SYSMIS)(999=SYSMIS). DO IF (SEX=1) AND (ABCC99 LE 102). COMPUTE WCBIN99=0. ELSE IF (SEX=1) AND (ABCC99 GT 102). COMPUTE WCBIN99=1. END IF. DO IF (SEX=2) AND (ABCC99 LE 88). COMPUTE WCBIN99=0. ELSE IF (SEX=2) AND (ABCC99 GT 88). COMPUTE WCBIN99=1. END IF. RECODE WCBIN99 (SYSMIS=999).

SHYP992M

Purpose : High systolic blood pressure at age 53yrs Year : 1999

1. Definition

0 No (<130mmHg and no hypertensive medication use)
1 Yes (≥130mmHg and no hypertensive medication use)
999 Missing

2. Specification

DO IF (B02299A=0) AND (B02499A=0) AND (B02599A=0) AND (B026299A=0). COMPUTE MEDBP99=0. ELSE IF (B02299A=1) OR (B02499A=1) OR (B02599A=1) OR (B026299A=1). COMPUTE MEDBP99=1. END IF.

RECODE SBP992 (9999=SYSMIS). RECODE DBP992 (9999=SYSMIS).

DO IF (SBP992 LT 130) AND (MEDBP99=0). COMPUTE SHYP992M=0. ELSE IF (SBP992 GE 130) OR (MEDBP99=1). COMPUTE SHYP992M=1. END IF. RECODE SHYP992M (SYSMIS=999).

DHYP992M

Purpose : High diastolic blood pressure at age 53yrs Year : 1999

1. Definition

0 No (<85 mmHg and no hypertensive medication use)
1 Yes (≥85mmHg and no hypertensive medication use)
999 Missing

2. Specification

DO IF (DBP992 LT 85) AND (MEDBP99=0). COMPUTE DHYP992M=0. ELSE IF (DBP992 GE 85) OR (MEDBP99=1). COMPUTE DHYP992M=1. END IF. RECODE DHYP992M (SYSMIS=999).

HYP992M

Purpose : Hypertension or antihypertensive medication at age 53yrs Year : 1999

1. Definition

0 No
1 Yes 999 Missing

2. Specification

DO IF (SHYP992M EQ 1 OR DHYP992M EQ 1). COMPUTE HYP992M=1. ELSE IF (SHYP992M EQ 0 AND DHYP992M EQ 0). COMPUTE HYP992M=0. END IF. RECODE HYP992M (SYSMIS=999).

HBABIN99

Purpose : High level of glycated haemoglobin (using quartile as cut-off) at age 53yrs Year : 1999

1. Definition

0 No (≤ 5.8%) 1 Yes (>5.8%) 999 Missing

2. Specification

COMPUTE HBA99NT=HBA99N. RECODE HBA99NT (77=SYSMIS) (98=SYSMIS) (99=SYSMIS). RECODE HBA99NT (LO THRU 5.800=0) (5.800 THRU HI=1) INTO HBABIN99. RECODE HBABIN99 (SYSMIS=999).

HBABIN99M

Purpose : High level of glycated haemoglobin (using quartile as cut-off) or medication for diabetes at age 53yrs Year : 1999

1. Definition

0 No (≤ 5.8% and no medication for diabetes)
1 Yes (>5.8% or medication for diabetes)
999 Missing

2. Specification

DO IF (HBABIN99 EQ 0) AND (B06199A=0). COMPUTE HBABIN99M=0. ELSE IF (HBABIN99 EQ 1) OR (B06199A=1). COMPUTE HBABIN99M=1. END IF. RECODE HBABIN99M (SYSMIS=999).

HDLBIN99

Purpose : Low level of high density lipoprotein at age 53yrs Year : 1999

1. Definition

0 No
1 Yes 999 Missing

2. Specification

COMPUTE HDL99NT=HDL99N. RECODE HDL99NT (77=SYSMIS) (98=SYSMIS) (99=SYSMIS). DO IF (SEX=1) AND (HDL99NT GE 1.036). COMPUTE HDLBIN99=0. ELSE IF (SEX=1) AND (HDL99NT LT 1.036). COMPUTE HDLBIN99=1. END IF. DO IF (SEX=2) AND (HDL99NT GE 1.295). COMPUTE HDLBIN99=0. ELSE IF (SEX=2) AND (HDL99NT LT 1.295). COMPUTE HDLBIN99=1. END IF. RECODE HDLBIN99 (SYSMIS=999).

TRIGBIN99

Purpose : High level of triglyceride at age 53yrs using ATPIII criteria Year : 1999

1. Definition

0 No
1 Yes 999 Missing

2. Specification

** Using 1.7 mmol/L as a cut-off (according ATPIII)

COMPUTE TRIG99NT=TRIG99N. RECODE TRIG99NT (77=SYSMIS) (98=SYSMIS) (99=SYSMIS). DO IF (TRIG99NT LT 1.7). COMPUTE TRIGBIN99=0. ELSE IF (TRIG99NT GE 1.7). COMPUTE TRIGBIN99=1. END IF. RECODE TRIGBIN99 (SYSMIS=999).

TRIGBIN99X

Purpose : High level of triglyceride at age 53yrs using quartile as a cut-off Year : 1999

1. Definition

0 No
1 Yes 999 Missing

2. Specification

**Using 2.6 mmol/L as a cut-off (quartile)

DO IF (SEX=1) AND (TRIG99NT LT 3.0). COMPUTE TRIGBIN99X=0. ELSE IF (TRIG99NT GE 3.0). COMPUTE TRIGBIN99X=1. END IF. DO IF (SEX=2) AND (TRIG99NT LT 2.2). COMPUTE TRIGBIN99X=0. ELSE IF (TRIG99NT GE 2.2). COMPUTE TRIGBIN99X=1. END IF. RECODE TRIGBIN99X (SYSMIS=999).

MSSCORE99

Purpose : Number of metabolic syndrome symptoms at age 53yrs Year : 1999

1. Definition

0 None
1-5 Valid score 999 Missing

2. Specification

**Using high level of triglycerides as defined by ATPIII (>=1.7mmol/L)

RECODE WCBIN99 HYP992M HBABIN99M HDLBIN99 TRIGBIN99 (999=SYSMIS). COMPUTE MSSCORE99 = WCBIN99+HYP992M+HBABIN99M+HDLBIN99+TRIGBIN99.

MSSCORE99X

Purpose : Number of metabolic syndrome symptoms at age 53yrs (with TRIGBIN99X) Year : 1999

1. Definition

0 None
1-5 Valid score 999 Missing

2. Specification

**Using high level of triglycerides as defined by us with quartile as a cu-off(>=2.6mmol/L)

RECODE WCBIN99 HYP992M HBABIN99M HDLBIN99 TRIGBIN99X (999=SYSMIS). COMPUTE MSSCORE99X = WCBIN99+HYP992M+HBABIN99M+HDLBIN99+TRIGBIN99X.

METABOL99

Purpose : Metabolic syndrome at age 53yrs Year : 1999

1. Definition

0 No (<3 symptoms presented)
1 Yes (≥3 symptoms presented) 999 Missing

2. Specification

**Using high level of triglycerides as defined by ATPIII (>=1.7mmol/L)

DO IF (MSSCORE99 GE 3). COMPUTE METABOL99=1. ELSE IF (MSSCORE99 LT 3). COMPUTE METABOL99=0. END IF. RECODE MSSCORE99 METABOL99 (SYSMIS=999).

METABOL99X

Purpose : Metabolic syndrome at age 53yrs (with TRIGBIN99X instead of TRIGBIN99) Year : 1999

1. Definition

0 No (<3 symptoms presented)
1 Yes (≥3 symptoms presented) 999 Missing

2. Specification

**Using high level of triglycerides as defined by us with quartile as a cu-off(>=2.6mmol/L)

DO IF (MSSCORE99X GE 3). COMPUTE METABOL99X=1. ELSE IF (MSSCORE99X LT 3). COMPUTE METABOL99X=0. END IF. RECODE MSSCORE99X METABOL99X (SYSMIS=999).

Variables in this document

14 high-confidence matches

Variable	Label	How it was found
Biological samples [52] › Blood assays [201] › Blood biochemistry [2011]		
hbabin99	High level of glycated haemoglobin (No/Yes). Using > 5.8% cut-off for both men and women - at age 53 years	topsheet field
hbabin99m	High level of glycated haemoglobin (No/Yes). Using > 5.8% cut-off for both men and women or medication for diabetes - at age 53 years	topsheet field
hdlbin99	Low level of high density lipoprotein (No/Yes). HDL cholesterol level cut-off was < 1.036 mmol/L for men or <1.295 mmol/L for women - at age 53 years	topsheet field
metabol99	Metabolic syndrome (with TRIGBIN) - at age 53 years	topsheet field
metabol99x	Metabolic syndrome (with TRIGBINX) - at age 53 years	topsheet field
msscore99	Number of metabolic syndrome symptoms (with TRIGBIN) - at age 53 years	topsheet field
msscore99x	Number of metabolic syndrome symptoms (with TRIGBINX) - at age 53 years	topsheet field
trigbin99	High level of triglyceride (No/Yes). Using 1.7 mmol/L as a cut-off (according ATPIII) - at age 53 years	topsheet field
trigbin99x	High level of triglyceride (No/Yes). Using ≥ 3.0 mmol/L for men and ≥ 2.2 for women as a cut-off (quartile). - at age 53 years	topsheet field
Clinical data [51] › Physical measures [110] › Anthropometry [1102] › Body size measures [11021]		
wcbin99	High waist circumference (No/Yes). High waist circumference is defined as: > 102 cm for men or > 88 cm for women - at age 53 years	topsheet field
Clinical data [51] › Physical measures [110] › Blood pressure [1103]		
dhyp992m	High diastolic blood pressure (No/Yes). High DBP: DBP ≥85mmHg or antihypertensive medication use - at age 53 years	topsheet field
shyp992m	High systolic blood pressure (No/Yes) High SBP: SBP ≥ 130 or antihypertensive medications use. - at age 53 years	topsheet field
Questionnaire data [55] › Health and medical history [510] › Medical conditions [5101] › Endocrine and metabolic conditions [51015]		
diabms	Diabetes status (No/Yes) - at age 53 years	topsheet field

Variable	Label	How it was found
Questionnaire data [55] › Medications [505] › Regular medication [5052] › Regular medication - Derived variables [50522] › Medication derivations at data collection level (cross-sectional) [505221]		
hyp992m	Hypertension or antihypertensive medication use (No/Yes) - at age 53 years	topsheet field

18 medium-confidence matches

Variable	Label	How it was found
Biological samples [52] › Blood assays [201] › Blood biochemistry [2011]		
chol99n	Cholesterol (mmol/l) - at age 53 years	prose scan
hba99n	Glycohaemoglobin - HbA1c (%) - at age 53 years	prose scan
hdl99n	High-density lipoprotein - HDL (mmol/L) at age 53 years	prose scan
ldl99n	Low-density lipoproteins - LDL (mmol/L) at age 53 years	prose scan
trig99n	Triglcyeride (mmol/l) - at age 53 years	prose scan
Clinical data [51] › Physical measures [110] › Anthropometry [1102] › Body size measures [11021]		
bmi99u	Body mass index (BMI) in kg/(m2) at 53 years	prose scan
Clinical data [51] › Physical measures [110] › Blood pressure [1103]		
dbp992	Diastolic blood pressure (mmHg) - second reading - at age 53 years	prose scan
sbp992	Systolic blood pressure (mmHg) - second reading - at age 53 years	prose scan
Clinical data [51] › Physical measures [110] › Hand examination [1106]		
problems	Any problems identifed with the hands - Hands Examinations 1999	prose scan
Questionnaire data [55] › Health and medical history [510] › Medical conditions [5101] › Endocrine and metabolic conditions [51015]		
diabty	Type of diabetes - up to age 53 years	prose scan
Questionnaire data [55] › Health and medical history [510] › Medical conditions [5101] › Other medical conditions [51011]		
blood	Have you had anaemia in the last 10 years - at age 43 years	prose scan
Questionnaire data [55] › Medications [505] › Regular medication [5052] › Regular medication - Derived variables [50522] › Medication derivations at data collection level (cross-sectional) [505221]		
b02299a	Antihypertensive medication (diuretics) - at age 53 years	prose scan
b02499a	Antihypertensive medication (beta blockers) - at age 53 years	prose scan
b02599a	Antihypertensive medication (RA system drugs and others) - at age 53 years	prose scan
b026299a	Antihypertensive medication (calcium channel blockers) - at age 53 years	prose scan
b06199a	Derived: Regular medication - BNFs 061xxx - drugs used in diabetes - BNF 6.1 at age 53 years	prose scan
b212099a	Derived: Regular medication - BNFs 2120xx - lipid-regulating drugs (BNF 40) - lipid-lowering drugs (BNF 27) - BNF 2.12 at age 53 years	prose scan
medbp99	Does participant take any antihypertensive medication at age 53 years	prose scan

1 low-confidence match

Variable	Label	How it was found
Population characteristics [54] › Baseline characteristics [401]		
sex	Sex of the participant	prose scan